Software Testing Life Cycle

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- It is a sequence of testing phases or activities that ensure testing is organized and complete.
- Testing activities;
  - Requirement analysis
  - Test planning
  - Test case design
  - Test environment set up
  - Test execution
  - Defect reporting and tracking
  - Retesting
  - Regression testing
  - Test closure

1. Requirement Analysis

- Involves understanding what needs to be tested i.e. functional or non-functional requirements.
- **FMS Example**
  - Review requirements to ensure that income is split correctly into salaries, operations, food, development and loans.

2. Test Planning

- Define the scope, approach, resources, schedule and tools for testing.
- **FMS Example**
  - Plan to do unit, integration, regression, performance tests for income distribution module.

3. Test Case Design

- Involves creating detailed test cases with inputs, expected outputs and execution steps.
- **FMS Example**
  - Enter UGX100,000 to check if income distribution percentages are correct.
  - Generate daily income report to verify that totals match.

4. Test Environment Set up

- Prepare hardware, software and data for testing.
- **FMS Example**
  - Set up Excel sheets, databases etc.

5. Test Execution

- Running test cases and comparing actual with expected results.
- **FMS Example**
  - Enter daily income
  - Generate daily income distribution report

6. Defect Reporting and Tracking

- Recording defects found during testing and tracking them until they are fixed.
- **FMS Example**
  - Bug: “Income distribution totals 99% instead of 100% ”

7. Retesting

- Testing again after defects are fixed to ensure the fixes work correctly.
- **FMS Example**
  - Verify that the corrected percentage calculations sums to 100%.

8. Regression Testing

- Re-run previously passed test cases to ensure new changes have not broken existing functionality.
- **FMS Example**
  - Ensure adding a new category “Transport (5%)” does not affect the existing income distribution calculations.

9. Test Closure

- Evaluating testing, preparing reports and documenting lessons learnt.
- **FMS Example**
  - Summarize test results
  - Document issues for future improvements e.g. upgrade Excel system to web application
- Sign off: Approval that testing was sufficient, complete and system is ready for deployment.

Test Plan

- A test plan is a formal document that describes the scope, approach, resources, tools, schedule of testing activities.
- Components: test plan ID, introduction, scope, objectives, approach etc.

1. Test Plan ID

- This is a unique identifier for the plan.
- **FMS Example**
  - FMS-TP-001

2. Introduction

- A brief description of the system and testing objectives.
- **FMS Example**
  - This plan covers functional and non-functional testing of FMS to ensure accurate income and expense tracking.

3. Scope

- Shows what will and will not be tested.
- **FMS Example**
  - Will test income entry module, income distribution, expenses and reports
  - Will NOT test payroll integration

4. Objectives

- Goals for testing
- **FMS Example**
  - To verify accurate balances, correct reports, system performance and security.

5. Testing Strategy or Approach

- High level plan that describes how testing will be done. Includes method, types, technique of testing and tools required.
- **FMS Example**
  - Use black box testing to verify income distribution
  - Use manual testing for user interface
  - Perform unit testing for individual modules

6. Test Items

- Specifies functions or modules to test.
- **FMS Example**
  - Income entry module, income distribution module, expense module, reporting module

7. Test Environment

- Describes hardware and software tools.
- **FMS Example**
  - Excel sheets, Windows 10 PC

8. Test Data

- Data sets used for testing
- **FMS Example**
  - Daily income of UGX100,000
  - Expenses in different categories
  - Extreme and negative input values

9. Schedule

- This is the timeline for testing activities.
- **FMS Example**

Testing activity	Time
Unit testing	2 – 4 April
Integration	10 – 12 April

10. Roles and Responsibilities

- Defines who does what
- **FMS Example**

Role	Name	Responsibility
Tester	John	Executes test cases
Developer	Faith	Fixes defects

11. Entry/Exit Criteria

- Defines conditions met before testing starts or completes.
- **FMS Example**
  - Entry: All modules coded, test data
  - Exit: All critical defects fixed, all planned test cases have been executed, system is ready

12. Deliverables

- Outputs from testing
- **FMS Example**
  - Test cases, defect reports, test summary report, sign off

13. Risk and Mitigation

- Potential problems and solutions
- **FMS Example**
  - Risk: High number of transactions may slow system
  - Mitigation: ...

14. Approval

- Sign-off from authority to indicate testing is sufficient, complete and system is ready for use.
- QA manager, project lead, client or end user

Test Plan - FMS Example

- **Objective**: Ensure FMS distributes income into salaries, operations, food, development and loans correctly.
- **Test Items**: Income entry module, distribution module, expense module, reporting module
- **Test data**: UGX1,000,000 daily income, expense UGX50,000
- **Schedule**: Unit, integration and system testing
- **Entry Criteria:** Test cases reviewed and approved
- **Exit criteria**: All income distribution tests pass, reports match as expected, no critical defects

Assignment 2

- Prepare a detailed test plan for your application (5 marks).

Test Case

- A test case is a documented set of inputs, conditions and expected results used to verify that a software works correctly.
- Test case design ensures complete testing coverage, provides documentation, helps find defects early.
- A good test case should be clear, simple, reusable and linked to requirements.

Components of a Test Case

Test case ID	Unique identifier
Test Scenario	What is being tested
Preconditions	What must be true before testing
Test steps	Actions to perform
Test Data	Input values
Expected results	Correct output
Actual result	What happened during testing
Status	Pass/Fail
Remarks	Additional information or observations

Test Case – FMS Login

Test case ID	TC_LOGIN_01
Test Scenario	Login with valid credentials
Preconditions	User account exists
Test steps	1. Enter username 2. Enter password 3. Click Login
Test Data	Username: admin Password: 1234
Expected results	Successful login
Actual result	Successful login
Status	…
Remarks	…

Types of Test Cases

Functional	Verify system functions
Negative	Use invalid inputs
Boundary	Test limits
Usability	Check user-friendliness
Performance	Check speed and response
Security	Check data protection

Assignment 3

- Write 10 test cases for your application (5 marks)

END